MINISTRY OF EDUCATION AND VOCATIONAL TRAINING

CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

URBAN WEST REGION - ZANZIBAR

FORM FOUR (IV) MOCK EXAMINATION - 2025

CHEMISTRY 2A

(For Both School and Private Candidates)

032/2A **(ACTUAL PRACTICAL)**

TIME: 2:30 HOURS JUNE 2025

INSTRUCTIONS:

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **25** marks
3. Cellular phones and any unauthorized materials are **Not allowed** in the examination room
4. Mathematical tables and non-programmable calculators **may be used**.
5. Write your **examination number** on every page of your answer booklet(s).
6. For calculations you **may use** the following constant

- Atomic masses: H = 1, O =16, Na = 23, Cl = 35.5,C = 12,S=32,K=39.

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1. A large volume of water has been contaminated with sulphuric acid .Before the contaminated water can be discarded off; the acid must be neutralized by adding potassium carbonate. You are required to determine the concentration of the acid in contaminated water and then calculating the moles of potassium carbonate needed to neutralize all the acid.

Solution **L** is a sample of contaminated water.
Solution **P** is sample made by dissolving 5.6g of potassium hydroxide in 1 dm^3 of solution

Solution **MO** is methyl orange indicator

QUESTIONS

(a) Titrate **L**(in burette) against **10 cm^3** of **P** (in conical flask) using two drops of indicator and obtained three titre values. Record your data in a tabular form.

(b) (i) Find the mean titre volume.
(ii) Write the summary of the experiment
(iii) What was the colour change during your experiment?

(c) (i) Calculate the molarity of the analyte .
(ii) From a balanced chemical equation, calculate the molarity of the contaminated water.
(iii) Find the number of moles of sulphuric acid present in 5000 dm^3 of contaminated water.

(d) From a balanced chemical equation, calculate the number of moles of potassium carbonate needed to neutralized all the acid in 5000 dm^3 ofcontaminated water.

2. Substance **Y** is a simply salt which contains one cation and one anion. Carry out the experiments describe below and record carefully your observations to make appropriate inferences and hence identify the cation and anion present in sample **Y.**

S/N	EXPERIMENT	OBSERVATION	INFERENCE
1	Observe the appearance of sample Y		
2	Perform a flame test of sample Y.		
3	Heat a little amount of sample Y in a dry test tube.		
4	Prepare stock solution and divide the resulting solution into three portions and add the following:		
	(a) To the first portion, add NaOH solution till excess.		
	(b) To the second portion add few drops of barium chloride solution followed by dilute hydrochloric acid.		
5	To the third portion perform confirmatory test for cation.		

Conclusion:

(i) The cation in sample **Y** is _____ and the anion is _____
(ii) The chemical formula of sample **Y** is _____
(iii) The name of compound **Y** is _____